Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/15/2021
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight,Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/22/2016	**End Date:**	06/21/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/21/2021 per NSSC information (Ed., 8/7/19)		
Key Personnel Changes/Previous PI:	January 2018 report: Prof. Alain Karma is no longer a Co-Investigator on this project, but continues to be a collaborator, who is responsible for the modeling part of the project through a separate NASA grant.		
COI Name (Institution):			
Grant/Contract No.:	NNX16AL77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)" grant number NNX12AK54G, with the same Principal Investigator Dr. Rohit Trivedi. During the manufacturing of industrial components by solidification processes, which include casting, welding, and single crystal growth, the properties and the reliability of the resulting products depend on the microstructure, which is controlled by the alloy composition and the processing conditions of growth rate and thermal gradient at the solidifying interface. It is thus crucial to quantitatively relate the critical microstructural features with the growth conditions and the composition of the alloy so that one can select alloy composition and design growth conditions to obtain microstructure that gives optimum properties to the finished product. From a fundamental standpoint, the formation of different solidification microstructures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced convection in the liquid under terrestrial conditions. Therefore, directional solidification experiments are designed on International Space Station (ISS) to investigate microstructure development under negligible convection to obtain benchmark data that can be used to validate the predictions of theoretical models that are also developed in this project. The most common solid-liquid interface patterns that form during the directional solidification processing of metals consist of spatially extended arrays of cells or dendrites. In previous microgravity experiments, i.e., Directional Solidification Insert (DSI) experiments, we examined the time evolution of spatially extended three-dimensional patterns under a wide range of velocity for two different thermal gradients in succinonitrile-0.24 wt% camphor. For this dilute composition, the microstructures were mainly cellular. In the current DSI-R project experiments were carried out on the International Space Station in a higher composition alloy, succinonitrile- 0.5 wt % camphor, in which dendritic arrays formed at lower velocities where the macroscopic curvature of the interface was minimized. The dynamics of 3D dendritic arrays were quantitatively examined and primary spacing variation with velocity was examined. Primary spacing was found to depend on the orientation of dendrites with respect to the heat flow direction. We have competed experiments on primary spacing evolution and the next step will be to study the evolution of secondary dendrite branches that play a critical role in determining properties of the solidifies alloy. Since only one experimental set was carried out on International Space Station, whose results will be analyzed after receiving the images, the efforts in the current reporting period were concentrated on analyzing available data from DSI experiments on oscillating cellular array and on ground-based experiments on side branch evolution.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based super alloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process that is used to fabricate appropriate components. These microstructures in turn control the integrity and mechanical properties of cast parts. By combining well-controlled and accurately characterized experimental evolution of three-dimensional solidification microstructures with quantitative phase-field simulations, this research should bring significant progress in understanding the origin and propagation of side-branching and the nature of the cell to dendrite transition, which have been a long standing issue in solidification for several decades. The deeper understanding of microstructure formation dynamics, including the development of side-branched structure in the mushy zone, shall thus accelerate and optimize the design of new technological alloys and their processing conditions that should directly benefit US industries.
Task Progress:	REPORTING PERIOD: JUNE 2019-JUNE 2020 One of the exciting observations in our previous results was the discovery of a novel oscillatory pattern in a cellular array in which three groups of cells were found to synchronize into oscillatory breathing modes. These 3D oscillating patterns were analyzed by characterizing the variation in cell area with time for a group of cells. This characterization of oscillating cells was not complete as changes in cell tip shape and the position of the cell tip were not measurable from these pictures of the interface taken from the top camera. However, the necessary information was in interferometric images taken during growth. We analyzed these interferometric as well as optical images to reconstruct the cell shape, position of each cell tip, tip radius, and tip velocity during its oscillation cycle. These results show a definite phase relationship. The cell tip radii and the cell areas are generally in phase and well correlated. This is because both these variables describe the same phenomena, i.e., an increase of the tip radius corresponds to a larger and flatter tip that is associated with an increase in surface area, and they can be considered to describe the lateral stretching/shrinking of cells during the oscillating cycle. Another important observation on the oscillating cells observed in images was the effect of misoriented cells on the oscillating phenomenon. Previous observations and analyses were carried out on a cellular array in which the preferred growth direction of cells was aligned with the heat flow direction. However, the sample was melted and regrown several times on ISS, which produced many sub-boundaries. The presence of sub-boundaries indicated a small tilting of the cells such that the orientation of cells in each sub-grain was initially at some angle with respect to the heat flow direction, and the cell growth direction changed during growth to align with the preferred crystallographic direction. As the cell direction changes with growth, there is a drift of cells and this drift velocity was measured from the images of cells obtained from the top camera. The drift induced by growth misorientation was found to influence cell oscillations, and from the analysis of experimental data we proposed a criterion for oscillations to stop when the ratio of characteristic times for drift and oscillation was between 2.3 and 2.8. Ground-based studies in thin samples were carried out to characterize the conditions for incoherent to coherent dendrite side branch formation. We first examined the effect of thermal gradient and primary spacing on side branch formation. When the thermal gradient was increased, the first instability appeared closer to the tip and a value of thermal gradient was found when the side branch formed closer enough to the tip to interact with the tip. With larger thermal gradients, the side branch formation mechanism was observed to change from noise induced to limit cycle. The effect of primary spacing was found such that at larger spacing a lower thermal gradient was required for the transition. We thus conclude that a larger thermal gradient and a larger primary spacing favor the formation of coherent side branching. Results of all experiments with varying thermal gradient and primary spacing will now be analyzed to obtain a quantitative criterion.
Bibliography Type:	Description: (Last Updated: 04/03/2024)

Abstracts for Journals and Proceedings	Mota FL, Pereda J, Song Y, Ji K, Tourret D, Strutzenberg L, Trivedi R, Karma A, Bergeon N. "3D pattern adjustment during directional solidification of a transparent alloy conducted on DECLIC-DSI." 5th International Conference on Advances in Solidification Processes, ICASP, Salzberg, Austria, June 17-21, 2019. Abstracts. 5th International Conference on Advances in Solidification Processes, ICASP, Salzberg, Austria, June 17-21, 2019. , Jun-2019
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